

Instructions:

- The assignment consists of **three questions**. Each question carries **equal weightage**. The **total marks for the assignment is 90**.
- Students must submit:
 - A **well-commented, runnable simulation code** (either **MATLAB** or **Python**).
 - A **short report** containing the required plots, figures, and a clear interpretation of the obtained results.
- The report should clearly explain the simulation approach, assumptions (if any), and the physical interpretation of the results.
- **Submission Deadline: 31 March 2026.**

1. 2D Simulation of a Parallel-Plate Capacitor with Finite Dimensions

Develop a numerical 2D electrostatic simulation to compute the electric potential $V(x, y)$ and the electric field $\mathbf{E}(x, y)$ for a parallel-plate capacitor with finite plate dimensions. Use a discretized square computational domain and solve Laplace's equation iteratively.

- **Computational domain:** A square region of size $5 \text{ mm} \times 5 \text{ mm}$ discretized into an $N \times N$ grid with $N = 120$.
- **Top plate (Plate A):** Fixed potential $V = +5 \text{ V}$, extending from $x = -2.5 \text{ mm}$ to $x = 2.5 \text{ mm}$ at $y = +\frac{5}{3} \text{ mm}$.
- **Bottom plate (Plate B):** Fixed potential $V = -5 \text{ V}$, extending from $x = -2.5 \text{ mm}$ to $x = 2.5 \text{ mm}$ at $y = -\frac{5}{3} \text{ mm}$.
- **Medium:** Assume a uniform dielectric with relative permittivity $\kappa = 1$ (vacuum).
- **Optional extension:** Introduce a dielectric interface:
 - Region $y > 0$: $\kappa_1 = 4$
 - Region $y < 0$: $\kappa_2 = 1$
- **Boundary condition:** All outer boundaries of the computational domain are grounded ($V = 0$).
- **Iteration scheme:** Solve Laplace's equation using the nearest-neighbor averaging method for at least 1000 iterations.

Required outputs

- Plot the electric field lines within the simulation domain.
- Plot the equipotential contour map of $V(x, y)$.

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- Plot the variation of the electric field magnitude along the vertical line passing through the midpoint of the two plates.

2. 2D Simulation of a Point Charge Near a Grounded Conducting Sphere

Develop a numerical simulation to study the electrostatic interaction between a point charge and a grounded conducting sphere. Investigate how the induced charge distribution on the sphere changes with the position and magnitude of the external charge.

- **Conducting sphere:** Centered at the origin with radius $R = 2$ (in simulation units) and held at $V = 0$.
- **Point charge:** $Q = 10 \mu\text{C}$ placed at $(x, y) = (4, 0)$.
- **Simulation region:** Square region defined by $-6 \leq x \leq 6$ and $-6 \leq y \leq 6$.
- Perform additional simulations by:
 - Changing the magnitude of the point charge.
 - Changing the distance of the point charge from the sphere.

Required outputs

- Plot the induced surface charge density on the sphere.
- Plot the equipotential contours in the simulation region.
- Plot the electric field vector distribution in the simulation region.

3. Simulation of Electric Field Enhancement at a Sharp Conductor (Lightning Rod)

Simulate the electrostatic field distribution produced by a sharp conducting needle placed between two conducting planes representing ground and a charged cloud.

- Use a 100×100 computational grid.
- **Bottom boundary (ground):** Set the entire bottom row to $V = 0$.
- **Top boundary (cloud):** Set the entire top row to $V = 100$.
- **Needle conductor:** At the center of the grid, create a vertical conducting line extending from the ground up to the middle of the grid and fix it at $V = 0$.
- Solve Laplace's equation iteratively until the potential distribution converges.

Required outputs

- Generate a heat map (color map) of the potential distribution.
- Overlay the electric field vectors computed using $\mathbf{E} = -\nabla V$.
- Describe the electric field behavior near the tip of the needle and explain the physical reason for this effect.